

Q1. Let A be a non-singular matrix of order 3. If $\det(3 \operatorname{adj}(2 \operatorname{adj}((\det A)A))) = 3^{-13} \cdot 2^{-10}$ and $\det(3 \operatorname{adj}(2A)) = 2^m \cdot 3^n$, then $|3m + 2n|$ is equal to 09 Apr 2024 (M)

Q2. Let $B = \begin{bmatrix} 1 & 3 \\ 1 & 5 \end{bmatrix}$ and A be a 2×2 matrix such that $AB^{-1} = A^{-1}$. If $BCB^{-1} = A$ and $C^4 + \alpha C^2 + \beta I = O$, then $2\beta - \alpha$ is equal to 09 Apr 2024 (E)

(1) 16

(2) 2

(3) 8

(4) 10

Q3. Let $A = \begin{bmatrix} 2 & a & 0 \\ 1 & 3 & 1 \\ 0 & 5 & b \end{bmatrix}$. If $A^3 = 4A^2 - A - 21I$, where I is the identity matrix of order 3×3 , then $2a + 3b$ is equal to 08 Apr 2024 (M)

(1) -9

(2) -13

(3) -10

(4) -12

Q4. Let $A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$. If the sum of the diagonal elements of A^{13} is 3^n , then n is equal to 08 Apr 2024 (M)

Q5. If A is a square matrix of order 3 such that $\det(A) = 3$ and $\det(\operatorname{adj}(-4 \operatorname{adj}(-3 \operatorname{adj}(3 \operatorname{adj}((2A)^{-1})))) = 2^m 3^n$, then $m + 2n$ is equal to : 06 Apr 2024 (E)

(1) 2

(2) 3

(3) 6

(4) 4

Q6. Let $\alpha\beta \neq 0$ and $A = \begin{bmatrix} \beta & \alpha & 3 \\ \alpha & \alpha & \beta \\ -\beta & \alpha & 2\alpha \end{bmatrix}$. If $B = \begin{bmatrix} 3\alpha & -9 & 3\alpha \\ -\alpha & 7 & -2\alpha \\ -2\alpha & 5 & -2\beta \end{bmatrix}$ is the matrix of cofactors of the elements of A , then $\det(AB)$ is equal to : 05 Apr 2024 (E)

(1) 64

(2) 216

(3) 343

(4) 125

Q7. Let A and B be two square matrices of order 3 such that $|A| = 3$ and $|B| = 2$. Then $|A^T A (\operatorname{adj}(2A))^{-1} (\operatorname{adj}(4B)) (\operatorname{adj}(AB))^{-1} A A^T|$ is equal to : 05 Apr 2024 (M)

(1) 108

(2) 32

(3) 81

(4) 64

Q8. Let A be a 3×3 matrix of non-negative real elements such that $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Then the maximum value of $\det(A)$ is 04 Apr 2024 (M)

Q9. Let A be a 2×2 symmetric matrix such that $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ and the determinant of A be 1. If $A^{-1} = \alpha A + \beta I$, where I is an identity matrix of order 2×2 , then $\alpha + \beta$ equals 04 Apr 2024 (E)

Q10. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = I + \text{adj}(A) + (\text{adj } A)^2 + \dots + (\text{adj } A)^{10}$. Then, the sum of all the elements of the matrix B is:

04 Apr 2024 (E)

(1) -124

(2) 22

(3) -88

(4) -110

Q11. Let $\alpha \in (0, \infty)$ and $A = \begin{bmatrix} 1 & 2 & \alpha \\ 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$. If $\det(\text{adj}(2A - A^T) \cdot \text{adj}(A - 2A^T)) = 2^8$, then $(\det(A))^2$ is equal to:

04 Apr 2024 (M)

(1) 36

(2) 16

(3) 1

(4) 49

Q12. If $A = \begin{bmatrix} \sqrt{2} & 1 \\ -1 & \sqrt{2} \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $C = ABA^T$ and $X = A^T C^2 A$, then $\det X$ is equal to:

01 Feb 2024 (M)

(1) 243

(2) 729

(3) 27

(4) 891

Q13. Let $A = I_2 - 2MM^T$, where M is real matrix of order 2×1 such that the relation $M^T M = I_1$ holds. If λ is a real number such that the relation $AX = \lambda X$ holds for some non-zero real matrix X of order 2×1 , then the sum of squares of all possible values of λ is equal to:

01 Feb 2024 (E)

Q14. Let A be a 3×3 matrix and $\det(A) = 2$. If $n = \det \left(\underbrace{\text{adj}(\text{adj}(\dots(\text{adj } A)))}_{2024\text{-times}} \right)$, then the remainder when n is divided by 9 is equal to _____.

31 Jan 2024 (E)

Q15. Let $R = \begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{pmatrix}$ be a non-zero 3×3 matrix, where $x \sin \theta = y \sin \left(\theta + \frac{2\pi}{3} \right) = z \sin \left(\theta + \frac{4\pi}{3} \right) \neq 0$, $\theta \in (0, 2\pi)$. For a square matrix M , let $\text{Trace}(M)$ denote the sum of all the diagonal entries of M . Then, among the statements:

(I) $\text{Trace}(R) = 0$ (II) If $\text{Trace}(\text{adj}(\text{adj}(R))) = 0$, then R has exactly one non-zero entry.

30 Jan 2024 (E)

(1) Both (I) and (II) are true

(2) Only (II) is true

(3) Neither (I) nor (II) is true

(4) Only (I) is true

Q16. Let $A = \begin{bmatrix} 2 & 1 & 2 \\ 6 & 2 & 11 \\ 3 & 3 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 2 & 0 \\ 5 & 0 & 2 \\ 7 & 1 & 5 \end{bmatrix}$. The sum of the prime factors of $|P^{-1}AP - 2I|$ is equal to

29 Jan 2024 (E)

(1) 26

(2) 27

(3) 66

(4) 23

Q17. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \beta & \alpha \end{bmatrix}$ and $|2A|^3 = 2^{21}$ where $\alpha, \beta \in \mathbb{Z}$, Then a value of α is

29 Jan 2024 (M)

- (1) 3 (2) 5
(3) 17 (4) 9

Q18. Let A be a square matrix such that $AA^T = I$. Then $\frac{1}{2} A \left[(A + A^T)^2 + (A - A^T)^2 \right]$ is equal to

29 Jan 2024 (M)

- (1) $A^2 + I$ (2) $A^3 + I$
(3) $A^2 + A^T$ (4) $A^3 + A^T$

Q19. Let A be a 2×2 real matrix and I be the identity matrix of order 2. If the roots of the equation $|A - xI| = 0$ be -1 and 3 , then the sum of the diagonal elements of the matrix A^2 is _____.

27 Jan 2024 (E)

Q20. Let $A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$, $B = [B_1 \ B_2 \ B_3]$, where B_1, B_2, B_3 are column matrices, and $AB_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$,

$$AB_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}, AB_3 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

If $\alpha = |B|$ and β is the sum of all the diagonal elements of B , then $\alpha^3 + \beta^3$ is equal to

27 Jan 2024 (M)

Q21. Consider the matrix $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Given below are two statements :

Statement I: $f(-x)$ is the inverse of the matrix $f(x)$.

Statement II: $f(x)f(y) = f(x+y)$.

In the light of the above statements, choose the correct answer from the options given below 27 Jan 2024 (M)

- (1) Statement I is false but Statement II is true (2) Both Statement I and Statement II are false
(3) Statement I is true but Statement II is false (4) Both Statement I and Statement II are true

Q22. Let the determinant of a square matrix A of order m be $m - n$, where m and n satisfy $4m + n = 22$ and

$17m + 4n = 93$. If $\det(n \operatorname{adj}(\operatorname{adj}(mA))) = 3^a 5^b 6^c$, then $a + b + c$ is equal to

15 Apr 2023 (M)

- (1) 84 (2) 96
(3) 101 (4) 109

Q23. The number of symmetric matrices of order 3, with all the entries from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is

13 Apr 2023 (M)

- (1) 6^{10} (2) 10^6
(3) 9^{10} (4) 10^9

Q24. Let $B = \begin{bmatrix} 1 & 3 & \alpha \\ 1 & 2 & 3 \\ \alpha & \alpha & 4 \end{bmatrix}$, $\alpha > 2$ be the adjoint of a matrix A and $|A| = 2$. Then $[\alpha \ -2\alpha \ \alpha]B \begin{bmatrix} \alpha \\ -2\alpha \\ \alpha \end{bmatrix}$ is equal to

13 Apr 2023 (M)

- (1) 0
(3) -16

- (2) 16
(4) 32

Q25.

Let for $A = \begin{bmatrix} 1 & 2 & 3 \\ \alpha & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}$, $|A| = 2$. If $|2 \operatorname{adj} (2 \operatorname{adj} (2A))| = 32^n$, then $3n + \alpha$ is equal to

- (1) 9
(3) 12

- (2) 11
(4) 10

13 Apr 2023 (E)

Q26.

Let $A = \begin{bmatrix} 1 & \frac{1}{51} \\ 0 & 1 \end{bmatrix}$. If $B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} A \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}$, then the sum of all the elements of the matrix $\sum_{n=1}^{50} B^n$ is equal to

- (1) 75
(3) 50

- (2) 125
(4) 100

12 Apr 2023 (M)

Q27.

Let $A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$, where $a, c \in \mathbb{R}$. If $A^3 = A$ and the positive value of a belongs to the interval $(n-1, n]$, where $n \in \mathbb{N}$, then n is equal to _____.

11 Apr 2023 (M)

Q28. Let A be a 2×2 matrix with real entries such that $A' = \alpha A + 1$, where $\alpha \in \mathbb{R} - \{-1, 1\}$. If $\det (A^2 - A) = 4$, the sum of all possible values of α is equal to

11 Apr 2023 (M)

- (1) 0
(3) 2

- (2) $\frac{3}{2}$
(4) $\frac{5}{2}$

Q29. If A is a 3×3 matrix and $|A| = 2$, then $|3 \operatorname{adj} (|3A|A^2)|$ is equal to

10 Apr 2023 (M)

- (1) $3^{12} \cdot 6^{11}$
(3) $3^{10} \cdot 6^{11}$

- (2) $3^{12} \cdot 6^{10}$
(4) $3^{11} \cdot 6^{10}$

Q30.

If $A = \frac{1}{5!6!7!} \begin{bmatrix} 5! & 6! & 7! \\ 6! & 7! & 8! \\ 7! & 8! & 9! \end{bmatrix}$, then $|\operatorname{adj} (\operatorname{adj} (2A))|$ is equal to

10 Apr 2023 (E)

- (1) 2^{20}
(3) 2^{12}

- (2) 2^8
(4) 2^{16}

Q31.

If $A = \begin{bmatrix} 1 & 5 \\ \lambda & 10 \end{bmatrix}$, $A^{-1} = \alpha A + \beta I$ and $\alpha + \beta = -2$, then $4\alpha^2 + \beta^2 + \lambda^2$ is equal to :

08 Apr 2023 (E)

- (1) 12
(3) 14

- (2) 19
(4) 10

Q32.

Let $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$. If $P^T Q^{2007} P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $2a + b - 3c - 4d$ is equal to

08 Apr 2023 (M)

- (1) 2004
(3) 2007

- (2) 2005
(4) 2006

Q33. Let $\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$. If $|\text{adj}(\text{adj}(\text{adj} 2A))| = (16)^n$, then n is equal to

08 Apr 2023 (M)

- (1) 8 (2) 10
(3) 9 (4) 12

Q34. Let $A = [a_{ij}]_{2 \times 2}$, where $a_{ij} \neq 0$ for all i, j and $A^2 = I$, Let a be the sum of all diagonal elements of A and $b = |A|$ Then $3a^2 + 4b^2$ is equal to

06 Apr 2023 (M)

- (1) 4 (2) 14
(3) 7 (4) 3

Q35. Let P be a square matrix such that $P^2 = I - P$. For $\alpha, \beta, \gamma, \delta \in \mathbb{N}$, if $P^\alpha + P^\beta = \gamma I - 29P$ and $P^\alpha - P^\beta = \delta I - 13P$, then $\alpha + \beta + \gamma - \delta$ is equal to

06 Apr 2023 (E)

- (1) 18 (2) 40
(3) 22 (4) 24

Q36. If $A = \frac{1}{2} \begin{bmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}$ then,

01 Feb 2023 (E)

- (1) $A^{30} - A^{25} = 2I$ (2) $A^{30} + A^{25} + A = I$
(3) $A^{30} + A^{25} - A = I$ (4) $A^{30} = A^{25}$

Q37. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{bmatrix}$. Then the sum of the diagonal elements of the matrix $(A + I)^{11}$ is equal to:

31 Jan 2023 (M)

- (1) 6144 (2) 4094
(3) 4097 (4) 2050

Q38. Let A be a $n \times n$ matrix such that $|A| = 2$. If the determinant of the matrix $\text{Adj} (2 \cdot \text{Adj} (2 A^{-1}))$ is 2^{84} , then n is equal to _____.

31 Jan 2023 (E)

Q39. Let $A = [a_{ij}]$, $a_{ij} \in \mathbb{Z} \cap [0, 4]$, $1 \leq i, j \leq 2$. The number of matrices A such that the sum of all entries is a prime number $p \in (2, 13)$ is _____.

31 Jan 2023 (E)

Q40. If P is a 3×3 real matrix such that $P^T = aP + (a - 1)I$, where $a > 1$, then

30 Jan 2023 (E)

- (1) P is a singular matrix (2) $|\text{Adj } P| > 1$
(3) $|\text{Adj } P| = \frac{1}{2}$ (4) $|\text{Adj } P| = 1$

Q41. Let $A = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$, $d = |A| \neq 0$ and $|A - d(\text{Adj } A)| = 0$. Then

30 Jan 2023 (M)

- (1) $(1 + d)^2 = (m + q)^2$ (2) $1 + d^2 = (m + q)^2$
(3) $(1 + d)^2 = m^2 + q^2$ (4) $1 + d^2 = m^2 + q^2$

Q42. Let α and β be real numbers. Consider a 3×3 matrix A such that $A^2 = 3A + \alpha I$. If $A^4 = 21A + \beta I$, then

29 Jan 2023 (M)

(1) $\alpha = 1$

(3) $\beta = 8$

(2) $\alpha = 4$

(4) $\beta = -8$

Q43. Let A be a symmetric matrix such that $|A| = 2$ and $\begin{bmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{bmatrix} A = \begin{bmatrix} 1 & 2 \\ \alpha & \beta \end{bmatrix}$. If the sum of the diagonal elements of A is s , then $\frac{\beta s}{\alpha^2}$ is equal to _____.

29 Jan 2023 (E)

Q44. The set of all values of $t \in \mathbb{R}$, for which the matrix $\begin{bmatrix} e^t & e^{-t}(\sin t - 2 \cos t) & e^{-t}(-2 \sin t - \cos t) \\ e^t & e^{-t}(2 \sin t + \cos t) & e^{-t}(\sin t - 2 \cos t) \\ e^t & e^{-t} \cos t & e^{-t} \sin t \end{bmatrix}$ is invertible, is

29 Jan 2023 (E)

(1) $\{(2k+1)\frac{\pi}{2}, k \in \mathbb{Z}\}$

(3) $\{k\pi, k \in \mathbb{Z}\}$

(2) $\{k\pi + \frac{\pi}{4}, k \in \mathbb{Z}\}$

(4) $\mathbb{R}$

Q45. Let A, B, C be 3×3 matrices such that A is symmetric and B and C are skew-symmetric.

Consider the statements

(S1) $A^{13} B^{26} - B^{26} A^{13}$ is symmetric(S2) $A^{26} C^{13} - C^{13} A^{26}$ is symmetric

Then,

25 Jan 2023 (E)

(1) Only $S2$ is true(3) Both $S1$ and $S2$ are false(2) Only $S1$ is true(4) Both $S1$ and $S2$ are true

Q46. Let $A = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \\ -\frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -i \\ 0 & 1 \end{bmatrix}$, where $i = \sqrt{-1}$. If $M = A^T B A$, then the inverse of the matrix $A M^{2023} A^T$ is

25 Jan 2023 (E)

(1) $\begin{bmatrix} 1 & -2023i \\ 0 & 1 \end{bmatrix}$

(3) $\begin{bmatrix} 1 & 0 \\ 2023i & 1 \end{bmatrix}$

(2) $\begin{bmatrix} 1 & 0 \\ -2023i & 1 \end{bmatrix}$

(4) $\begin{bmatrix} 1 & 2023i \\ 0 & 1 \end{bmatrix}$

Q47. Let $x, y, z > 1$ and $A = \begin{bmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 2 & \log_y z \\ \log_z x & \log_z y & 3 \end{bmatrix}$. Then $|\text{adj}(\text{adj } A^2)|$ is equal to

25 Jan 2023 (M)

(1) 6^4

(3) 4^8

(2) 2^8

(4) 2^4

Q48. If A and B are two non-zero $n \times n$ matrices such that $A^2 + B = A^2 B$, then

24 Jan 2023 (M)

(1) $AB = I$

(3) $A^2 = I$ or $B = I$

(2) $A^2 B = I$

(4) $A^2 B = BA^2$

Q49. The number of square matrices of order 5 with entries from the set $\{0, 1\}$, such that the sum of all the elements in each row is 1 and the sum of all the elements in each column is also 1, is

24 Jan 2023 (E)

(1) 225

(3) 150

(2) 120

(4) 125

Q50. Let A be a 3×3 matrix such that $|\text{adj}(\text{adj}(\text{adj}(A)))| = 12^4$. Then $|A^{-1}\text{adj} A|$ is equal to **24 Jan 2023 (E)**

(1) $2\sqrt{3}$

(3) 12

(2) $\sqrt{6}$

(4) 1

Q51. Which of the following matrices can NOT be obtained from the matrix $\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$ by a single elementary row operation? **29 Jul 2022 (E)**

(1) $\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$ (3) $\begin{bmatrix} -1 & 2 \\ -2 & 7 \end{bmatrix}$ (2) $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ (4) $\begin{bmatrix} -1 & 2 \\ -1 & 3 \end{bmatrix}$

Q52. Let $x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix}$. For $k \in \mathbb{N}$, if $X' A^k X = 33$, then k is equal to **29 Jul 2022 (E)**

Q53. The number of matrices of order 3×3 , whose entries are either 0 or 1 and the sum of all the entries is a prime number, is _____. **29 Jul 2022 (M)**

Q54. Let A and B be any two 3×3 symmetric and skew symmetric matrices respectively. Then which of the following is **NOT** true? **28 Jul 2022 (E)**

(1) $A^4 - B^4$ is a symmetric matrix(3) $B^5 - A^5$ is a skew-symmetric matrix(2) $AB - BA$ is a symmetric matrix(4) $AB + BA$ is a skew-symmetric matrix

Q55. Let $A = \begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} \beta & 1 \\ 1 & 0 \end{bmatrix}$, $\alpha, \beta \in \mathbb{R}$. Let α_1 be the value of α which satisfies

$(A + B)^2 = A^2 + \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ and α_2 be the value of α which satisfies $(A + B)^2 = B^2$. Then $|\alpha_1 - \alpha_2|$ is equal to

28 Jul 2022 (M)

Q56. Let the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and the matrix $B_0 = A^{49} + 2A^{98}$. If $B_n = \text{Adj}(B_{n-1})$ for all $n \geq 1$, then

 $\det(B_4)$ is equal to**28 Jul 2022 (M)**(1) 3^{28} (3) 3^{32} (2) 3^{30} (4) 3^{36}

Q57. Let $A = \begin{pmatrix} 1 & 2 \\ -2 & -5 \end{pmatrix}$. Let $\alpha, \beta \in \mathbb{R}$ be such that $\alpha A^2 + \beta A = 2I$. Then $\alpha + \beta$ is equal to **27 Jul 2022 (M)**

(1) -10

(3) 6

(2) -6

(4) 10

Q58. Let $A = \begin{pmatrix} 4 & -2 \\ \alpha & \beta \end{pmatrix}$. If $A^2 + \gamma A + 18I = O$, then $\det(A)$ is equal to _____. **27 Jul 2022 (E)**

(1) -18 (2) 18 (3) -50 (4) 50

Q59. Let S be the set containing all 3×3 matrices with entries from $\{-1, 0, 1\}$. The total number of matrices

$A \in S$ such that the sum of all the diagonal elements of $A^T A$ is 6 is _____.

27 Jul 2022 (M)

Q60.

Consider a matrix $A = \begin{bmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta + \gamma & \gamma + \alpha & \alpha + \beta \end{bmatrix}$, where α, β, γ are three distinct natural numbers.

If $\frac{\det(\text{adj}(\text{adj}(\text{adj}(\text{adj}A))))}{(\alpha - \beta)^{16}(\beta - \gamma)^{16}(\gamma - \alpha)^{16}} = 2^{32} \times 3^{16}$, then the number of such 3- tuples (α, β, γ) is _____.

27 Jul 2022 (E)

Q61.

The number of matrices $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $a, b, c, d \in \{-1, 0, 1, 2, 3, \dots, 10\}$, such that $A = A^{-1}$, is _____.

26 Jul 2022 (E)

Q62.

Let $A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $B = \begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix}$, then the value of $A'BA$ is;

(1) 1224

(2) 1042

(3) 540

(4) 539

26 Jul 2022 (E)

Q63. Let A be a 2×2 matrix with $\det(A) = -1$ and $\det((A + I)(\text{Adj}(A) + I)) = 4$. Then the sum of the diagonal elements of A can be:

(1) -1

(2) 2

(3) 1

(4) $-\sqrt{2}$

26 Jul 2022 (M)

Q64.

Let $A = \begin{bmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$ and $B = A - I$. If $\omega = \frac{\sqrt{3}i-1}{2}$, then the number of elements in the set

$\{n \in \{1, 2, \dots, 100\} : A^n + (\omega B)^n = A + B\}$ is equal to _____.

25 Jul 2022 (M)

Q65.

Let $A = \begin{bmatrix} 1 & a & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}$, $a, b \in \mathbb{R}$. If for some $n \in \mathbb{N}$, $A^n = \begin{bmatrix} 1 & 48 & 2160 \\ 0 & 1 & 96 \\ 0 & 0 & 1 \end{bmatrix}$ then $n + a + b$ is equal to _____.

25 Jul 2022 (E)

Q66. Let $A = [a_{ij}]$ be a square matrix of order 3 such that $a_{ij} = 2^{j-i}$, for all $i, j = 1, 2, 3$. Then, the matrix

$A^2 + A^3 + \dots + A^{10}$ is equal to

29 Jun 2022 (M)

(1) $\left(\frac{3^{10}-1}{2}\right)A$

(2) $\left(\frac{3^{10}+1}{2}\right)A$

(3) $\left(\frac{3^{10}+3}{2}\right)A$

(4) $\left(\frac{3^{10}-3}{2}\right)A$

Q67.

Let $M = \begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix}$, where α is a non-zero real number and $N = \sum_{k=1}^{49} M^{2k}$. If $(I - M^2)N = -2I$, then the positive integral value of α is _____.

29 Jun 2022 (E)

Q68. Let $A = \begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}$. If $B = I - {}^5C_1(\text{adj } A) + {}^5C_2(\text{adj } A)^2 - \dots - {}^5C_5(\text{adj } A)^5$, then the sum of

all elements of the matrix B is:

29 Jun 2022 (E)

(1) -5

(2) -6

(3) -7

(4) -8

Q69. Let $A = \begin{pmatrix} 1+i & 1 \\ -i & 0 \end{pmatrix}$ where $i = \sqrt{-1}$. Then, the number of elements in the set $\{n \in \{1, 2, \dots, 100\} : A^n = A\}$ is

28 Jun 2022 (E)

Q70. Let A be a matrix of order 3×3 and $\det(A) = 2$. Then $\det(\det(A) \text{adj}(5 \text{adj}(A^3)))$ is equal to _____.

28 Jun 2022 (M)

(1) 256×10^6

(2) 1024×10^6

(3) 512×10^6

(4) 256×10^{11}

Q71. Let A and B be two 3×3 matrices such that $AB = I$ and $|A| = \frac{1}{8}$ then $|\text{adj}(B \text{adj}(2A))|$ is equal to

27 Jun 2022 (E)

(1) 128

(2) 32

(3) 64

(4) 102

Q72.

The positive value of the determinant of the matrix A , whose $\text{Adj}(\text{Adj}(A)) = \begin{bmatrix} 14 & 28 & -14 \\ -14 & 14 & 28 \\ 28 & -14 & 14 \end{bmatrix}$, is _____.

27 Jun 2022 (M)

Q73. Let A be a matrix of order 2×2 , whose entries are from the set $\{0, 1, 2, 3, 4, 5\}$. If the sum of all the entries of A is a prime number p , $2 < p < 8$, then the number of such matrices A is

27 Jun 2022 (E)

Q74. Let A be a 3×3 invertible matrix. If $|\text{adj}(24A)| = |\text{adj}(3 \text{adj}(2A))|$, then $|A|^2$ is equal to

26 Jun 2022 (M)

(1) 2^6

(2) 2^{12}

(3) 512

(4) 6^6

Q75.

Let $X = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $Y = \alpha I + \beta X + \gamma X^2$ and $Z = \alpha^2 I - \alpha \beta X + (\beta^2 - \alpha \gamma) X^2$, $\alpha, \beta, \gamma \in \mathbb{R}$.

If $Y^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{-2}{5} & \frac{1}{5} \\ 0 & \frac{1}{5} & \frac{-2}{5} \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$, then $(\alpha - \beta + \gamma)^2$ is equal to _____.

26 Jun 2022 (E)

Q76. Let A be a 3×3 matrix having entries from the set $\{-1, 0, 1\}$. The number of all such matrices A having sum of all the entries equal to 5, is _____

25 Jun 2022 (M)

Q77. Let $A = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$. If M and N are two matrices given by $M = \sum_{k=1}^{10} A^{2k}$ and $N = \sum_{k=1}^{10} A^{2k-1}$ then MN^2

is

25 Jun 2022 (M)

- (1) a non-identity symmetric matrix (2) a skew-symmetric matrix
(3) neither symmetric nor skew-symmetric matrix (4) an identity matrix

Q78. Let $A = \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}$. Then the number of elements in the set $\{$

$(n, m) : n, m \in \{1, 2, \dots, 10\}$ and $nA^n + mB^m = I\}$ is _____. **25 Jun 2022 (E)**

Q79. Let $S = \left\{ \begin{pmatrix} -1 & a \\ 0 & b \end{pmatrix} ; a, b \in \{1, 2, 3, \dots, 100\} \right\}$ and let $T_n = \{A \in S : A^{n(n+1)} = I\}$. Then the number of elements in $\bigcap_{n=1}^{100} T_n$ is _____. **24 Jun 2022 (E)**

Q80. Let $S = \{\sqrt{n} : 1 \leq n \leq 50 \text{ and } n \text{ is odd}\}$. Let $a \in S$ and $A = \begin{bmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$. If $\sum_{a \in S} \det(\text{adj } A) = 100\lambda$, then λ

is equal to

- (1) 218 (2) 221
(3) 663 (4) 1717

Q81. Let $J_{n,m} = \int_0^{1/2} \frac{x^n}{x^m - 1} dx, \forall n > m$ and $n, m \in N$. Consider a matrix $A = [a_{ij}]_{3 \times 3}$ where

$a_{ij} = \begin{cases} J_{6+i,3} - J_{i+3,3} & , i \leq j \\ 0 & , i > j \end{cases}$. Then $|\text{adj } A^{-1}|$ is :

- (1) $(15)^2 \times 2^{34}$ (2) $(15)^2 \times 2^{42}$
(3) $(105)^2 \times 2^{36}$ (4) $(105)^2 \times 2^{38}$

Q82. The number of elements in the set $\{A = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix} : a, b, d \in \{-1, 0, 1\} \text{ and } (I - A)^3 = I - A^3\}$, where I is 2×2 identity matrix, is _____. **31 Aug 2021 (E)**

Q83. If the matrix $A = \begin{bmatrix} 0 & 2 \\ K & -1 \end{bmatrix}$ satisfies $A(A^3 + 3I) = 2I$, then the value of K is

- (1) $\frac{1}{2}$ (2) 1
(3) -1 (4) $-\frac{1}{2}$

Q84. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$. Then $A^{2025} - A^{2020}$ is equal to

- (1) $A^6 - A$ (2) A^6
(3) A^5 (4) $A^5 - A$

Q85. Let A be a 3×3 real matrix. If $\det(2 \text{ Adj}(2 \text{ Adj}(\text{Adj}(2A)))) = 2^{41}$, then the value of $\det(A^2)$ equals

_____. **26 Aug 2021 (E)**

Q86. If $A = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ i & 1 \end{bmatrix}$, $i = \sqrt{-1}$, and $Q = A^T B A$, then the inverse of the matrix $AQ^{2021}A^T$ is equal to:

26 Aug 2021 (M)

$$(1) \begin{bmatrix} 1 & -2021 \\ 0 & 1 \end{bmatrix}$$

$$(2) \begin{bmatrix} 1 & 0 \\ -2021i & 1 \end{bmatrix}$$

$$(3) \begin{bmatrix} \frac{1}{\sqrt{5}} & -2021 \\ 2021 & \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$(4) \begin{bmatrix} 1 & 0 \\ 2021 & i \end{bmatrix}$$

Q87. Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$. If $A^{-1} = \alpha I + \beta A$, $\alpha, \beta \in R$, I is a 2×2 identity matrix, then $4(\alpha - \beta)$ is equal to :

27 Jul 2021 (M)

$$(1) 5$$

$$(2) \frac{8}{3}$$

$$(3) 2$$

$$(4) 4$$

Q88. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and $M = A + A^2 + A^3 + \dots + A^{20}$, then the sum of all the elements of the matrix M is equal to _____.

27 Jul 2021 (E)

Q89. Let A and B be two 3×3 real matrices such that $(A^2 - B^2)$ is invertible matrix. If $A^5 = B^5$ and $A^3 B^2 = A^2 B^3$, then the value of the determinant of the matrix $A^3 + B^3$ is equal to :

27 Jul 2021 (E)

$$(1) 2$$

$$(2) 4$$

$$(3) 1$$

$$(4) 0$$

Q90. Let $S = \left\{ n \in N, \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \forall a, b, c, d \in R \right\}$, where $i = \sqrt{-1}$. Then the number of 2-digit numbers in the set S is _____.

25 Jul 2021 (M)

Q91. If $P = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$, then P^{50} is:

25 Jul 2021 (E)

$$(1) \begin{bmatrix} 1 & 0 \\ 25 & 1 \end{bmatrix}$$

$$(2) \begin{bmatrix} 1 & 50 \\ 0 & 1 \end{bmatrix}$$

$$(3) \begin{bmatrix} 1 & 25 \\ 0 & 1 \end{bmatrix}$$

$$(4) \begin{bmatrix} 1 & 0 \\ 50 & 1 \end{bmatrix}$$

Q92. Let $M = \left\{ A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in (\pm 3, \pm 2, \pm 1, 0) \right\}$. Define $f : M \rightarrow Z$, as $f(A) = \det(A)$, for all $A \in M$ where Z is set of all integers. Then the number of $A \in M$ such that $f(A) = 15$ is equal to _____.

25 Jul 2021 (M)

Q93. Let $A = [a_{ij}]$ be a real matrix of order 3×3 , such that $a_{i1} + a_{i2} + a_{i3} = 1$, for $i = 1, 2, 3$. Then, the sum of all the entries of the matrix A^3 is equal to:

22 Jul 2021 (M)

$$(1) 2$$

$$(2) 1$$

$$(3) 3$$

$$(4) 9$$

Q94. Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Then the number of 3×3 matrices B with entries from the set $\{1, 2, 3, 4, 5\}$ and satisfying $AB = BA$ is _____.

22 Jul 2021 (M)

Q95. Let $A = \{a_{ij}\}$ be a 3×3 matrix, where $a_{ij} = \begin{cases} (-1)^{j-i} & \text{if } i < j \\ 2 & \text{if } i = j \\ (-1)^{i+j} & \text{if } i > j \end{cases}$ then $\det(3 \operatorname{Adj}(2A^{-1}))$ is equal to _____.

20 Jul 2021 (E)

Q96. Let $A = \begin{bmatrix} 2 & 3 \\ a & 0 \end{bmatrix}$, $a \in R$ be written as $P + Q$ where P is a symmetric matrix and Q is skew symmetric matrix.

If $\det(Q) = 9$, then the modulus of the sum of all possible values of determinant of P is equal to:

20 Jul 2021 (M)

- (1) 36
(3) 45

- (2) 24
(4) 18

Q97. Let $A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$ and $B = 7A^{20} - 20A^7 + 2I$,

where I is an identity matrix of order 3×3 . If $B = [b_{ij}]$, then b_{13} is equal to

20 Jul 2021 (M)

Q98. Let $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$. If $\operatorname{Tr}(A)$ denotes the sum of all diagonal elements of the matrix A , then $\operatorname{Tr}(A) - \operatorname{Tr}(B)$ has value equal to

18 Mar 2021 (M)

- (1) 1
(3) 0

- (2) 2
(4) 3

Q99. Let I be an identity matrix of order 2×2 and $P = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$. Then the value of $n \in N$ for which

$P^n = 5I - 8P$ is equal to _____.

18 Mar 2021 (E)

Q100. If $A = \begin{bmatrix} 0 & \sin \alpha \\ \sin \alpha & 0 \end{bmatrix}$ and $\det(A^2 - \frac{1}{2}I) = 0$, then a possible value of α is

17 Mar 2021 (M)

- (1) $\frac{\pi}{2}$
(3) $\frac{\pi}{4}$

- (2) $\frac{\pi}{3}$
(4) $\frac{\pi}{6}$

Q101. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ such that $AB = B$ and $a + d = 2021$, then the value of $ad - bc$ is equal to _____.

17 Mar 2021 (E)

Q102. If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of $\det(A^4) + \det(A^{10} - (\operatorname{Adj}(2A))^{10})$ is equal to _____.

17 Mar 2021 (M)

Q103. Let $A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ be two 2×1 matrices with real entries such that $A = XB$, where

$X = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$, and $k \in R$. If $a_1^2 + a_2^2 = \frac{2}{3}(b_1^2 + b_2^2)$ and $(k^2 + 1)b_2^2 \neq -2b_1b_2$, then the value of k is

16 Mar 2021 (E)

Q104. The total number of 3×3 matrices A having entries from the set $(0, 1, 2, 3)$ such that the sum of all the diagonal entries of AA^T is 9, is equal to 16 Mar 2021 (M)

Q105. Let $P = \begin{bmatrix} -30 & 20 & 56 \\ 90 & 140 & 112 \\ 120 & 60 & 14 \end{bmatrix}$ and $A = \begin{bmatrix} 2 & 7 & \omega^2 \\ -1 & -\omega & 1 \\ 0 & -\omega & -\omega + 1 \end{bmatrix}$ where $\omega = \frac{-1+i\sqrt{3}}{2}$, and I_3 be the identity matrix of order 3. If the determinant of the matrix $(P^{-1}AP - I_3)^2$ is $\alpha\omega^2$, then the value of α is equal to 16 Mar 2021 (M)

Q106. Let A be a symmetric matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1, then the possible number of such matrices is: 26 Feb 2021 (M)

- (1) 12 (2) 4
(3) 1 (4) 6

Q107. If the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$ satisfies the equation $A^{20} + \alpha A^{19} + \beta A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ for some real numbers α and β , then $\beta - \alpha$ is equal to 26 Feb 2021 (E)

Q108. If for the matrix, $A = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}$, $AA^T = I_2$, then the value of $\alpha^4 + \beta^4$ is : 25 Feb 2021 (E)

- (1) 3 (2) 1
(3) 2 (4) 4

Q109. If $A = \begin{bmatrix} 0 & -\tan(\frac{\theta}{2}) \\ \tan(\frac{\theta}{2}) & 0 \end{bmatrix}$ and $(I_2 + A)(I_2 - A)^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then $13(a^2 + b^2)$ is equal to 25 Feb 2021 (M)

Q110. Let $A = \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$, where x , y and z are real numbers such that $x + y + z > 0$ and $xyz = 2$. If $A^2 = I_3$, then the value of $x^3 + y^3 + z^3$ is 25 Feb 2021 (M)

Q111. Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in R$. Suppose $Q = [q_{ij}]$ is a matrix satisfying $PQ = kI_3$ for some non-zero $k \in R$. If $q_{23} = -\frac{k}{8}$ and $|Q| = \frac{k^2}{2}$, then $\alpha^2 + k^2$ is equal to 24 Feb 2021 (M)

Q112. Let M be any 3×3 matrix with entries from the set $\{0, 1, 2\}$. The maximum number of such matrices, for which the sum of diagonal elements of $M^T M$ is seven, is 24 Feb 2021 (M)

Q113. Let $\theta = \frac{\pi}{5}$ and $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$. If $B = A + A^4$, then $\det(B)$: 06 Sep 2020 (E)

- (1) is one (2) lies in $(2, 3)$
(3) is zero (4) lies in $(1, 2)$

Q114. If $A = \begin{bmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{bmatrix}$, $(\theta = \frac{\pi}{24})$ and $A^5 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where $i = \sqrt{-1}$, then which one of the following is not true?

04 Sep 2020 (M)

(1) $0 \leq a^2 + b^2 \leq 1$

(2) $a^2 - d^2 = 0$

(3) $a^2 - c^2 = 1$

(4) $a^2 - b^2 = \frac{1}{2}$

Q115. Let $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$, $x \in R$ and $A^4 = [a_{ij}]$. If $a_{11} = 109$, then a_{22} is equal to _____.

03 Sep 2020 (M)

Q116. Let A be a 3×3 matrix such that $\text{adj } A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{bmatrix}$ and $B = \text{adj}(\text{adj } A)$. If $|A| = \lambda$ and

$|(B^{-1})^T| = \mu$, then the ordered pair $(|\lambda|, \mu)$ is equal to

03 Sep 2020 (E)

(1) $(3, \frac{1}{81})$

(2) $(9, \frac{1}{9})$

(3) $(3, 81)$

(4) $(9, \frac{1}{81})$

Q117. Let A be a 2×2 real matrix with entries from $\{0, 1\}$ and $|A| \neq 0$. Consider the following two statements;

(P) If $A \neq I_2$, then $|A| = -1$

(Q) If $|A| = 1$, then $\text{tr}(A) = 2$

Where I_2 denotes 2×2 identity matrix and $\text{tr}(A)$ denotes the sum of the diagonal entries of A . Then

02 Sep 2020 (M)

(1) (P) is false and (Q) is true

(2) Both (P) and (Q) are false

(3) (P) is true and (Q) is false

(4) Both (P) and (Q) are true

Q118. Let $A = \left\{ X = (x, y, z)^T : PX = 0 \text{ and } x^2 + y^2 + z^2 = 1 \right\}$ where $P = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 3 & -4 \\ 1 & 9 & -1 \end{bmatrix}$ then the set A

02 Sep 2020 (E)

(1) Is a singleton.

(2) Is an empty set.

(3) Contains more than two elements

(4) Contains exactly two elements

Q119. Let $a, b, c \in R$ be all non-zero and satisfies $a^3 + b^3 + c^3 = 2$. If the matrix $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ satisfies $A^T A = I$, then a value of abc can be

02 Sep 2020 (E)

(1) $-\frac{1}{3}$

(2) $\frac{1}{3}$

(3) 3

(4) $\frac{2}{3}$

Q120. If $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{bmatrix}$, $B = \text{adj } A$ and $C = 3A$, then $\frac{|\text{adj } B|}{|C|}$ is equal to

09 Jan 2020 (M)

(1) 8

(2) 16

(3) 72

(4) 2

Q121. If $A = \begin{pmatrix} 2 & 2 \\ 9 & 4 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, then $10 A^{-1}$, is equal to.

08 Jan 2020 (E)

(1) $A - 4I$

(2) $6I - A$

(3) $A - 6I$

(4) $4I - A$

Q122. The number of all 3×3 matrices A , with entries from the set $\{-1, 0, 1\}$ such that the sum of the diagonal elements of AA^T is 3, is _____.

08 Jan 2020 (M)

Q123.

Let α be a root of the equation $x^2 + x + 1 = 0$ and the matrix $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^4 \end{bmatrix}$, then the matrix A^{31} is

equal to

(1) A^3

(2) I_3

(3) A^2

(4) A

07 Jan 2020 (M)

Q124. If A is a symmetric matrix and B is skew-symmetric matrix such that $A + B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$, then AB is equal to:

12 Apr 2019 (M)

(1) $\begin{bmatrix} -4 & 2 \\ 1 & 4 \end{bmatrix}$

(2) $\begin{bmatrix} 4 & -2 \\ 1 & -4 \end{bmatrix}$

(3) $\begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$

(4) $\begin{bmatrix} -4 & -2 \\ -1 & 4 \end{bmatrix}$

Q125.

If $B = \begin{bmatrix} 5 & 2\alpha & 1 \\ 0 & 2 & 1 \\ \alpha & 3 & -1 \end{bmatrix}$ is the inverse of a 3×3 matrix A , then the sum of all values of α for which

 $\det(A) + 1 = 0$, is:

12 Apr 2019 (M)

(1) 2

(2) 1

(3) 0

(4) -1

Q126.

The total number of matrices $A = \begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}$, ($x, y \in \mathbb{R}$, $x \neq y$) for which $A^T A = 3I_3$ is:

09 Apr 2019 (E)

(1) 6

(2) 3

(3) 4

(4) 2

Q127. If $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$, then the inverse of $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is:

09 Apr 2019 (M)

(1) $\begin{bmatrix} 1 & -12 \\ 0 & 1 \end{bmatrix}$

(2) $\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}$

(3) $\begin{bmatrix} 1 & 0 \\ 13 & 1 \end{bmatrix}$

(4) $\begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$

Q128. Let $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$, ($\alpha \in \mathbb{R}$) such that $A^{32} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Then, a value of α is:

08 Apr 2019 (M)

(1) 0

(2) $\frac{\pi}{16}$

(3) $\frac{\pi}{64}$

(4) $\frac{\pi}{32}$

Q129. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $Q = [q_{ij}]$ be two 3×3 matrices such that $Q - P^5 = I_3$. Then $\frac{q_{21} + q_{31}}{q_{32}}$ is equal to :

12 Jan 2019 (M)

- (1) 10
(3) 15

- (2) 9
(4) 135

Q130. Let $A = \begin{pmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{pmatrix}$. If $AA^T = I_3$, then $|p|$ is:

11 Jan 2019 (M)

- (1) $\frac{1}{\sqrt{5}}$
(3) $\frac{1}{\sqrt{2}}$

- (2) $\frac{1}{\sqrt{3}}$
(4) $\frac{1}{\sqrt{6}}$

Q131. If $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, then the matrix A^{-50} when $\theta = \frac{\pi}{12}$, is equal to:

09 Jan 2019 (M)

- (1) $\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$
(3) $\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$

- (2) $\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$
(4) $\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

Q132. If $A = \begin{bmatrix} e^t & e^{-t} \cos t & e^{-t} \sin t \\ e^t & -e^{-t} \cos t - e^{-t} \sin t & -e^{-t} \sin t + e^{-t} \cos t \\ e^t & 2e^{-t} \sin t & -2e^{-t} \cos t \end{bmatrix}$, then A is:

09 Jan 2019 (E)

- (1) Invertible only if $t = \pi$
(3) Invertible only if $t = \frac{\pi}{2}$

- (2) Not invertible for any $t \in R$
(4) Invertible for all $t \in R$

ANSWER KEYS

1. (14)	2. (4)	3. (2)	4. (7)	5. (4)	6. (2)	7. (4)	8. (27)
9. (5)	10. (3)	11. (2)	12. (2)	13. (2)	14. (7)	15. (3)	16. (1)
17. (2)	18. (4)	19. (10)	20. (28)	21. (4)	22. (2)	23. (2)	24. (3)
25. (2)	26. (4)	27. (2)	28. (4)	29. (4)	30. (4)	31. (3)	32. (2)
33. (2)	34. (1)	35. (4)	36. (3)	37. (3)	38. (5)	39. (204)	40. (4)
41. (1)	42. (4)	43. (5)	44. (4)	45. (1)	46. (4)	47. (2)	48. (4)
49. (2)	50. (1)	51. (3)	52. (10)	53. (282)	54. (3)	55. (2)	56. (3)
57. (4)	58. (2)	59. (5376)	60. (42)	61. (50)	62. (4)	63. (2)	64. (17)
65. (24)	66. (4)	67. (1)	68. (3)	69. (25)	70. (3)	71. (3)	72. (14)
73. (180)	74. (1)	75. (100)	76. (414)	77. (1)	78. (1)	79. (100)	80. (2)
81. (4)	82. (8)	83. (1)	84. (1)	85. (4)	86. (2)	87. (4)	88. (2020)
89. (4)	90. (11)	91. (1)	92. (16)	93. (3)	94. (3125)	95. (108)	96. (1)
97. (910)	98. (2)	99. (6)	100. (3)	101. (2020)	102. (16)	103. (1)	104. (766)
105. (36)	106. (2)	107. (4)	108. (2)	109. (13)	110. (7)	111. (17)	112. (540)
113. (4)	114. (4)	115. (10)	116. (1)	117. (4)	118. (4)	119. (2)	120. (1)
121. (3)	122. (672)	123. (1)	124. (3)	125. (2)	126. (3)	127. (4)	128. (3)
129. (1)	130. (3)	131. (1)	132. (4)				

For solutions, [click here to download the MARKS App](#) from Google Play or visit web.getmarks.app on PC/desktop

